

Problem Set 1

Submission:

(Attention: Extended deadline!) Friday, 13/02/2026, until 13:30, HKT (UTC+08:00), to be uploaded on the Canvas course homepage.

Only problems marked with (*) will be graded.

1. Elementary combinatorics

Use the results on elementary combinatorics seen in the lecture to answer the following questions. Explain your reasoning.

- (a) How many possibilities are there to distribute 12 indistinguishable coins to 5 different persons?
- (b) Assume you have 9 colors to mark 5 different courses you have taken this term in your notes. How many possibilities to choose these colors are there, if every subject has its distinct color.
- (c) There are 10 lamps in a room, that can be turned on or off independently from each other. How many possibilities are there for the configuration of lights?
- (d) In a lottery, 6 out of 49 numbers are chosen at once. How many possibilities are there to guess exactly 3 out of these numbers correctly? Hence, calculate the probability to guess 3 out of 6 numbers correctly in this type of lottery.
- (e) There are 12 female and 8 male candidates for a committee made out of 8 persons in total. How many possibilities are there for the committee if there have to be at least 3 female and at most 5 male committee members?
- (f) On the unit circle, $n \geq 2$ equidistant points are put. Out of those, we choose two of them uniformly at random. Calculate the probability p_n that the chosen points are neighbors.
- (g) How many possible anagrams can be formed out of the name "HANNAH"? How many of those are palindromes (i.e. coinciding with the respective word read backwards)?
- (h) How many possibilities are there to order $n \geq 1$ books into a shelf, if $1 \leq m \leq n$ of the books are indistinguishable?

2. (*) Set operations and events, de Morgan's rules

[4 Points]

A (six-sided) die is rolled $N \in \mathbb{N}$ times. Let A_k be the event that the k -th role is a 3 and B_k is the event that the k -th roll is a 6 ($1 \leq k \leq N$).

- (a) Express with the help of the events A_k, B_k , as well as elementary set operations the following events:
 - i. A : 6 is never obtained.
 - ii. B : a 3 is obtained at least once.
 - iii. C : a 6 is obtained at least once and a 3 is obtained at least twice.
- (b) Describe the following events in words:

$$\left(\bigcup_{i=1}^N A_i^c \right)^c, \quad \bigcup_{i=1}^{N-2} (A_i \cap A_{i+1} \cap B_{i+2}), \quad \bigcap_{i=1}^{N-1} (A_i \cup B_{i+1}).$$

Now let $(A_i)_{i \in I}$ be a family of subsets of a non-empty set Ω . Prove *de Morgan's rules*:

$$(c) \left(\bigcup_{i \in I} A_i \right)^c = \bigcap_{i \in I} A_i^c.$$

$$(d) \left(\bigcap_{i \in I} A_i \right)^c = \bigcup_{i \in I} A_i^c.$$

3. (*) **Rolling dice**

[4 Points]

A fair die is rolled twice. Calculate the probability for the following events.

- (a) The number 6 occurs exactly once.
- (b) Both numbers are odd.
- (c) The sum of the numbers is 4.
- (d) The sum of the numbers is an integer multiple of 3.

Write down the probability space $(\Omega, \mathcal{F}, \mathbf{P})$ and the corresponding events explicitly.

4. **Inclusion-Exclusion formula**

Consider a probability space $(\Omega, \mathcal{F}, \mathbf{P})$. Let $(A_j)_{j=1}^n$ be events, $A_j \in \mathcal{F}$.

- (a) Using induction prove:

$$\mathbf{P} \left[\bigcup_{j=1}^n A_j \right] = \sum_{k=1}^n (-1)^{k+1} \sum_{1 \leq i_1 < \dots < i_k \leq n} \mathbf{P} \left[\bigcap_{j=1}^k A_{i_j} \right].$$

- (b) Using induction prove the following statements:

$$\mathbf{P} \left[\bigcup_{j=1}^n A_j \right] \leq \sum_{j=1}^n \mathbf{P}[A_j] - \sum_{j=1}^{n-1} \mathbf{P}[A_j \cap A_{j+1}],$$

$$\mathbf{P} \left[\bigcup_{j=1}^n A_j \right] \geq \sum_{j=1}^n \mathbf{P}[A_j] - \sum_{i,j=1, i < j}^n \mathbf{P}[A_j \cap A_i].$$